# Miscalibrated but Correctable: Bias in LLM-Based Digital Twin Demand Curves


## Abstract

Firms increasingly consider large language model (LLM)-based digital twins for simulating purchase decisions and estimating price-dependent purchase probabilities. This study tests whether a twin's aggregate demand curve can substitute for human stated-purchase data, using pricing decisions from 2,058 participants across 40 products in the Twin-2K-500 data set. The twin underpredicts purchase probability by 13 percentage points in the same direction across all 40 products. At the exact-price level, aggregation alone leaves twin error roughly twice the human test-retest benchmark, but on binned demand curves, a single calibration offset closes 57% of the gap to that benchmark.


## Statement of Key Contributions

This research contributes to marketing literature in three ways. First, it offers a demand-curve-level validation of an LLM-based digital twin against its matched human respondents, using human test-retest error rather than an externally imposed threshold as the benchmark. Second, it distinguishes aggregation from calibration: averaging reduces individual-level noise, whereas a global log-odds correction addresses the systematic level bias that remains. Third, a five-specification comparison shows that bias direction and magnitude vary by model and prompting method, while calibration reduces error in every specification tested. The relevant question is therefore not only whether twins are accurate, but whether their configuration-specific error is stable enough to correct. A secondary analysis finds no reliable evidence that subgroup demographic diversity predicts where twin error concentrates.

For non-academic stakeholders, the findings offer a concrete decision rule. Firms considering LLM-based digital twins for market sizing should not treat raw twin output as a substitute for human stated-purchase data. Much of the observed gap to human retest accuracy can be closed by applying a global correction estimated from human stated-purchase decisions, rather than estimating ground truth separately for every product. The required validation-sample size remains unknown, and the offset must be re-estimated when the model or prompting method changes. Exploratory subgroup and product patterns also warrant replication before informing operational decisions.

## *Introduction*

Large language models (LLMs) are increasingly used to build "digital twins" of survey respondents: AI personas that answer questions based on a human's prior survey responses (Toubia et al. 2025). While LLMs exhibit responses broadly consistent with some classic economic experiments (Horton, Filippas, and Manning 2023), direct tests of LLM-simulated willingness to pay show the resulting demand estimates can be inaccurate without correction (Brand, Israeli, and Ngwe 2023). Literature evaluating whether LLMs can substitute for human respondents shows mixed conclusions. Some report LLM responses track human behavior reasonably well (Argyle et al. 2023; Aher, Arriaga, and Kalai 2023), while others find systematic divergence from human data (Bisbee et al. 2024; Santurkar et al. 2023; Gui and Toubia 2023). Dillion et al. (2023) call for task-specific validation rather than general judgments about replacing human participants, while Bail (2024) raises concerns about training-data bias and reproducibility.

This study addresses a specific, managerially relevant version of that question. Can an LLM-based digital twin's aggregate demand curve, the relationship between price and purchase probability, be trusted as a stand-in for a human demand curve? Aggregate demand curves are a common input to pricing and market-sizing decisions, so their accuracy matters even when individual-level twin predictions are noisy. Prior work on LLM-simulated purchase intent (Maier et al. 2025; Li, Wei, and Wang 2025) has emphasized correlational fit with human ratings. The Twin-2K-500 data set paper reports that human wave-3 and wave-4 demand curves were practically indistinguishable and flagged a non-monotonic twin curve at zero prices (Toubia et al. 2025), but left the level of that bias unquantified and uncorrected. This study instead measures the twin's error directly against that same human benchmark and asks whether the bias is structured enough to be corrected.

Purchase decisions can be viewed as threshold judgments related to reservation price: a respondent buys when the product's perceived value is at least as high as its price. A digital twin must infer this judgment from incomplete information about the respondent's preferences. Its purchase threshold may therefore depend on the model, persona representation, and prompt. The twin may still capture how demand changes with price while consistently predicting too many or too few purchases. If so, its aggregate demand curve will mainly be shifted up or down rather than have a completely different shape. We therefore ask whether twin error is consistent with a shared threshold shift, whether one correction can transfer across products within a configuration, and whether the shift varies across model and prompting configurations.

Using pricing decisions from 2,058 participants across 40 consumer packaged goods in the Twin-2K-500 data set (Toubia et al. 2025), this study reports three findings. First, a GPT-4.1-mini digital twin under-predicts purchase probability in the same direction across all 40 products, and a global correction closes most, though not all, of the resulting gap. Second, bias direction varies across five model and prompting-method specifications, making calibration configuration specific. Third, demographic heterogeneity does not reliably predict twin error.[1]

## Data and Method

The Twin-2K-500 data set (Toubia et al. 2025) contains four waves of survey responses from 2,058 U.S. participants. A pricing module showed participants a randomly drawn price for each of 40 consumer packaged goods, spanning food, beverages, household supplies, pet products, and over-the-counter health products, and asked whether they would buy it. Wave 4 repeated the identical price for the same participant and product, providing a within-person retest. The wave-3-to-wave-4 disagreement rate therefore provides an empirical benchmark reflecting natural variability in repeated responses.

A GPT-4.1-mini digital twin, built from a text-based persona summarizing each participant's survey answers from waves prior to the pricing module, answered the same 40 pricing questions at the same randomly assigned prices; persona construction therefore did not use the wave-3/4 responses being predicted. All 2,058 participants had a usable twin response, giving one paired human-twin observation per participant, product, and price.

Throughout, "demand curve" refers to the aggregate relationship between assigned price and stated purchase probability, not realized market demand or quantity sold. Unless noted otherwise, twin error is measured against wave-3 responses, the same wave used to estimate the calibration offset. Twin accuracy was measured as the mean absolute error (MAE) between human and twin purchase probabilities. Two versions are reported. The exact-price version compares aggregate probabilities at each specific price shown. Because roughly 345 distinct prices were used per product across 2,058 people, each price was seen by about six people on average, so this metric retains substantial sampling noise. The binned version groups prices into 12 quantile-based bins per product before comparing probabilities, giving more observations per point and better isolating the demand curve's level; MAE weights each bin equally regardless of how many participants it contains. In short, the exact-price metric asks whether aggregation alone makes twins comparable to humans, while the binned metric asks whether calibration removes the systematic level difference that aggregation alone cannot.

The global probability offset $\omega$ was estimated as the paired human-minus-twin difference for each participant and product (wave 3), averaged first within each product and then equally across the 40 products, producing the reported 13.0-percentage-point aggregate gap. This offset was converted to an equivalent log-odds shift, anchored at each curve's own twin mean, and applied to the twin's binned demand curve:

$$p_{\text{corrected}} = \operatorname{logit}^{-1}\big(\operatorname{logit}(p_{\text{twin}}) + \delta\big), \qquad \delta = \operatorname{logit}(\bar{p}_{\text{twin}} + \omega) - \operatorname{logit}(\bar{p}_{\text{twin}})$$

[1] Code to reproduce all analyses is available at an anonymized repository: `https://anonymous.4open.science/r/markettwin-C134/`.

where $p_{\text{twin}}$ is the twin's purchase probability at a given price bin and $\bar{p}_{\text{twin}}$ is that curve's own twin mean, so the same probability-scale offset $\omega$ produces a log-odds shift $\delta$ specific to each curve. Probabilities were bounded away from 0 and 1 (clipped to $[10^{-6}, 1-10^{-6}]$) before the logit transform to avoid undefined values. This choice mattered: about 9% of correction points would otherwise have been pushed past the 0% or 100% boundaries and clipped back, flattening the curve's shape near the extremes, where ceiling and floor effects constrain the magnitude of possible adjustment. A second, more granular correction estimated a separate offset for each product rather than one shared value. Confidence intervals came from a non-parametric bootstrap: participants were resampled 500 times, using the same resampled index for the human and twin data so that each participant's pairing was preserved. Within each replication, the calibration offset and all downstream MAE and gap-closed statistics were re-estimated from scratch.

To test whether twin error is larger for demographically heterogeneous subgroups, each subgroup (for example, "income under $30,000" or "age 18 to 29") was given a heterogeneity score. The score averages how mixed that subgroup is on every other demographic variable, using normalized Gini impurity for both nominal and ordinal variables (missing values dropped per variable) and, for income and age, how wide that subgroup's own numeric range is. This score was regressed on twin MAE, controlling for subgroup size and product fixed effects, across four demographic variables: income, age, education, and political ideology.

## *Results*

### Aggregation helps, but does not close the gap

At the individual level, twin MAE was 0.329. This is consistent with the overall error rates reported by Toubia et al. (2025) across multiple tasks in that data set. At the aggregate level (exact-price grouping), twin MAE fell to 0.236, confirming that individual errors partly cancel when demand curves are aggregated across the sample.

The human retest MAE at the same aggregation level was 0.119, so aggregate twin error remained roughly twice as large. Aggregation narrows the difference between twin and human demand curves but does not close it. Figure 1 illustrates the persistent directional gap: for four representative products, the twin's curve sits below both human waves at nearly every price.

### A simple calibration closes most of the remaining gap

Averaged across all 40 products, humans' purchase probability was 43.1% in wave 3 and 44.1% in wave 4, nearly identical rates, as expected for a retest of the same people. The twin's purchase probability was 30.1%, 13.0 percentage points lower (bootstrap 95% CI: 12.2 to 13.7 points), and this under-prediction held in the same direction for all 40 products.

The uncorrected twin had an MAE of 0.122 on binned demand curves. A global log-odds intercept,

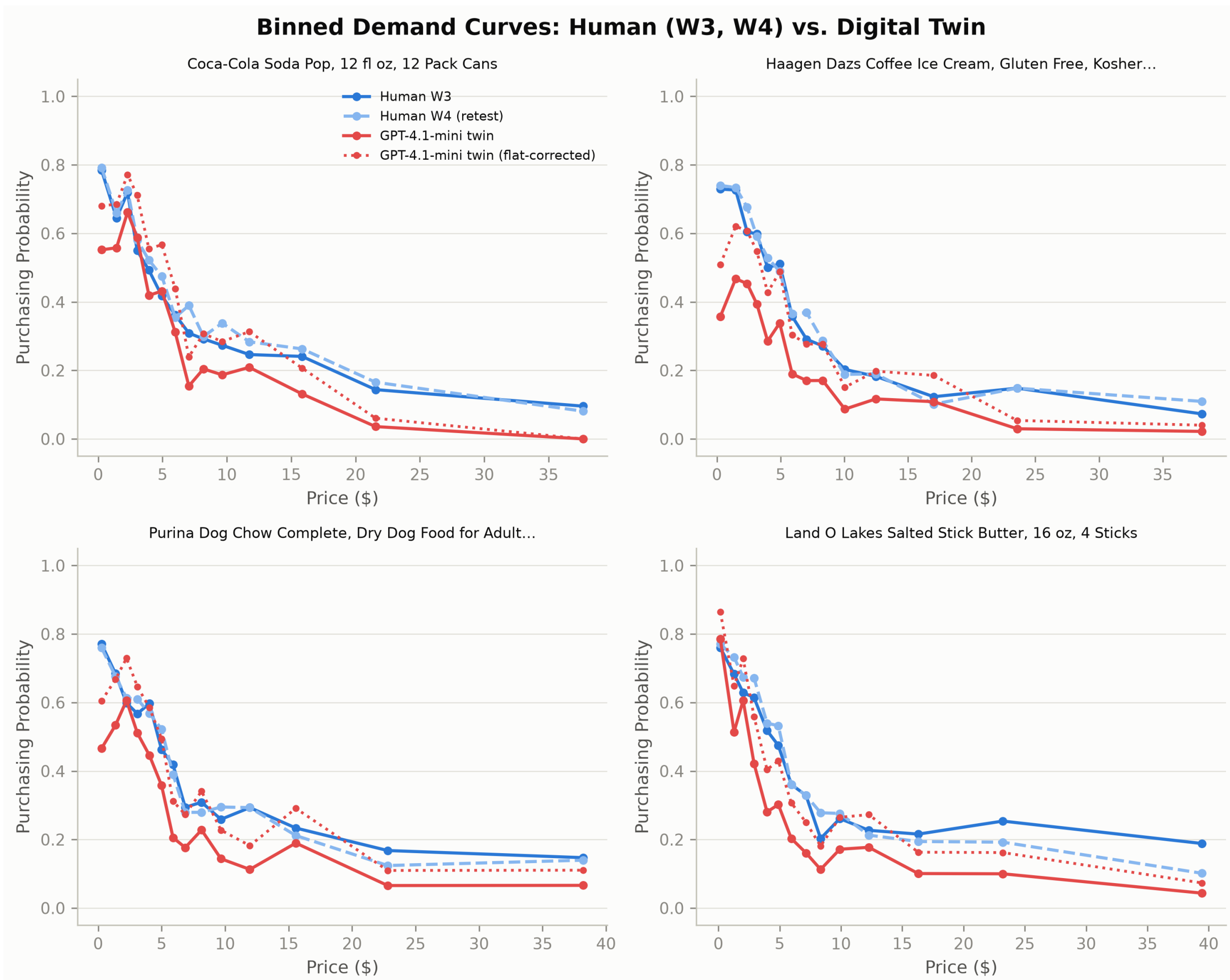


**Figure 1.** *Binned Demand Curves for Four Representative Products: Human (Waves 3 and 4) vs. Digital Twin, Raw and Globally Calibrated*

Each panel plots purchase probability against price for one product. The twin (solid red) sits below both human waves (solid and dashed blue) at most prices; the globally calibrated twin (dotted red) shifts back toward the human curves. The four products span the range of twin accuracy, from Coca-Cola Soda Pop (near the median) to Land O'Lakes Salted Stick Butter (the worst-calibrated of the 40 products).

calibrated to reconcile the 13.0-point aggregate probability gap, reduced MAE to 0.067, closing 57% of the gap to the human retest benchmark of 0.026 (bootstrap 95% CI: 49 to 58%). A leave-one-product-out check estimated each product's offset from the other 39 products and produced essentially the same result (MAE 0.067, 95% CI: 0.061 to 0.073; 57% closed), indicating within-category transfer. A separate offset for each product reduced MAE to 0.059 (65% closed, 95% CI: [0.055, 0.063]), but requires that product's human ground truth. The small interval overlap does not establish that this additional gain is statistically distinct without a paired test.

The global correction is more operationally feasible than per-product calibration because it uses shared validation data, although the required sample size is not established here. It is an average-case remedy: for 10 of 40 products, corrected MAE remained more than three times the human retest benchmark, potentially reflecting curve-shape mismatch or per-product sampling noise. Excluding $0-price observations changed aggregate error by less than half a percent. The dotted curves in Figure 1 show that the correction generally shifts the twin toward the human curves without directly estimating product-specific curve shapes; residual shape differences remain for some products.

### The direction of bias varies across configurations

The direction and magnitude of bias varied across the five specifications (Table 1). Four specifications over-predicted purchase probability, while the primary specification under-predicted it. For GPT-4.1-mini, changing the persona representation reversed the direction of bias. The model under-predicted purchase probability by 13.0 percentage points with a text persona but over-predicted it by 25.5 points with a JSON persona. Among the two JSON-persona specifications, GPT-4.1 had less bias than GPT-4.1-mini (5.6 versus 25.5 points), although the aligned samples differed. The fine-tuned specification had the highest aggregate error (MAE = 0.361), but this result does not identify the effect of fine-tuning because no matched baseline was available for its 1,558-person evaluation sample.

Table 1: Cross-Configuration Robustness Check: Bias and Calibration across Five Model and Prompting Specifications

| Model | Specification type | *N* | Offset (pp) | Direction | Exact-Price MAE | Binned Gap Closed |
|---|---|---|---|---|---|---|
| GPT-4.1-mini | Text persona (primary) | 2,058 | +13.0 | Under-predicts | 0.236 | 57% |
| GPT-4.1 | JSON persona | 2,031 | −5.6 | Over-predicts | 0.226 | 21% |
| GPT-4.1-mini | JSON persona | 1,896 | −25.5 | Over-predicts | 0.330 | 67% |
| Gemini-Flash2.5 | Text persona | 2,058 | −17.2 | Over-predicts | 0.274 | 56% |
| GPT-4.1-mini | Fine-tuned (500 samples) | 1,558 | −16.3 | Over-predicts | 0.361 | 30% |

*Note.* Offset is the signed difference between mean human and twin wave-3 purchase probabilities across 40 products (positive = under-prediction; negative = over-prediction). Exact-Price MAE is aggregate demand-curve MAE at exact prices. Binned Gap Closed is the share of the uncorrected-MAE-to-retest gap removed by global log-odds correction using 12 bins per product. *N* is the aligned sample for each specification. The fine-tuned model was trained on 500 participants and evaluated on 1,558. Exact-Price MAE and Binned Gap Closed use different aggregation bases and should not be compared within rows.

A global correction reduced MAE in all five specifications and closed between 21% and 67% of the gap to each specification's retest benchmark. Because the specifications were not evaluated on identical samples, these findings should be interpreted as a robustness pattern rather than as direct comparisons. Overall, the reversal within GPT-4.1-mini shows that miscalibration depends on the broader elicitation system, including persona representation, rather than on model identity alone.

### Subgroup diversity does not predict twin error

We tested whether twin error was greater for demographic subgroups whose members differed more on other personal characteristics. Twin MAE was regressed on the subgroup heterogeneity index, with controls for subgroup size and product fixed effects. Standard errors were clustered by product because subgroup observations within the same product were not independent. The analysis found no significant association between heterogeneity and twin error (coefficient $= -0.034$ MAE per unit of the heterogeneity index, 95% CI: $-0.133$ to 0.066, $p = 0.51$). The point estimate was negative, contrary to the hypothesized positive relationship, and the confidence interval included zero.

A broader analysis using all 14 available demographic variables, rather than the original four, also found no significant association (coefficient $= 0.066$, $p = 0.19$). The sign of the estimate changed across the two specifications, providing further evidence that the relationship was not stable.[2] Subgroup size was a significant predictor (coefficient $= -0.00004$, $p < .001$): larger subgroups had lower error. This pattern is consistent with sampling noise rather than an effect of demographic heterogeneity. Overall, demographic diversity did not reliably identify where digital twins were less accurate.

### Where twins fail most

Descriptive subgroup errors varied. The highest average error occurred among respondents who identified as very conservative (MAE = 0.347), followed by those with postgraduate education (MAE = 0.316) and those aged 18 to 29 (MAE = 0.309). The lowest occurred among respondents earning less than $30,000 (MAE = 0.262).

The political-ideology pattern is descriptively consistent with prior evidence that conservative opinions are comparatively underrepresented in human-feedback-tuned language models (Santurkar et al. 2023). The very conservative subgroup's excess error was positive for 38 of 40 products, but the analysis does not identify its source. Because subgroup comparisons were exploratory and not adjusted for multiple comparisons, this pattern requires replication before being interpreted as systematic miscalibration.

At the product level, error was highest for a small set of everyday grocery items: a butter product,

[2] This result uses a revised heterogeneity index that respects each demographic variable's measurement scale. An earlier index treated all demographic codes as continuous, including nominal categories, and produced a significant association that did not remain after this correction.

a Greek yogurt product, and a ramen noodle product, each among the highest MAE of the 40 products (0.26 to 0.27).

## *Discussion and Implications*

The main implication is that digital-twin accuracy depends on the overall simulation setup, not only on the language model. In the specifications examined here, both model choice and persona representation were associated with changes in the magnitude and direction of aggregate bias. Aggregation and calibration addressed different sources of error. Aggregation reduced individual-level noise, while calibration adjusted the systematic difference in average purchase probability. For the primary specification, the global correction closed most of the observed gap to the human retest benchmark. This result is consistent with a shared threshold shift accounting for an important part of the error. However, differences in curve shape and product-level accuracy remained. Calibration should therefore be judged against the intended decision. An average improvement may support broad market-sizing exercises, but it may be insufficient when decisions depend on a particular product or a narrow price range. The estimated offset also varied across configurations, so an offset from one configuration should not be assumed to apply to another. Each configuration therefore requires its own validation before its bias can be considered correctable. This finding builds on evidence that post hoc calibration can improve predictive accuracy (Guo et al. 2017; Platt 1999) and suggests that calibration should be evaluated within the specific human-twin simulation setup being considered.

For practice, firms should be cautious about using raw twin demand curves as substitutes for human stated-purchase data. Matched human validation data can help identify and partially correct systematic bias. A hybrid design that combines human and synthetic responses may therefore be more appropriate than complete substitution. This study does not establish how many human observations are needed to estimate a reliable correction. Demographic diversity also did not reliably identify which subgroups had the greatest error, so this pattern should be evaluated in each application. Changes to the model, persona representation, prompt structure, or fine-tuning procedure may also change the required offset. Validation should therefore be repeated when the simulation setup changes rather than treated as a one-time step.

## *Limitations and Future Research*

This study has several limitations. First, it uses one pregenerated set of responses per specification. The results therefore cannot separate model bias from prompting artifacts or run-to-run variation. The five-specification comparison provides a robustness check but does not resolve this issue, and the single fine-tuning run should not be treated as evidence about fine-tuning in general. Second, the human retest measured agreement at the same price rather than responses to new prices. The analysis also used 12 price bins per product and did not test other binning choices. Third, the study did not determine how many human observations are needed for calibration. The offset and gap-closed estimates both used wave-3 responses, while wave 4 was not used as an independent test of the offset. Fourth, all products were consumer packaged goods, so the leave-one-product-out analysis shows transfer only within this domain. The subgroup comparisons were not adjusted for multiple testing, making the political-ideology result suggestive rather than conclusive. Finally, the shared threshold shift is an interpretation, not an identified mechanism. Testing this explanation would require independent variation in model choice, persona representation, and prompt framing.

Covariate-aware calibration has been proposed for other LLM-alignment tasks (Maia Polo et al. 2025), but, to our knowledge, not for consumer-behavior simulation. The present results favor the global offset over more granular alternatives: per-product calibration closed only eight additional percentage points of the gap and required product-level human data, while demographics did not reliably predict error. This is consistent with evidence that corrected LLM-generated data may complement rather than replace human data (Wang, Zhang, and Zhang 2026). Future research should test the required validation-sample size, transfer across product categories and populations, subgroup-level calibration, alternative price bins, and persona prompts designed to reduce bias before calibration.

### *Conclusion*

An LLM-based digital twin's aggregate demand curve, used without adjustment, is not an adequate out-of-the-box substitute for human stated-purchase data: its exact-price error was roughly twice the human retest benchmark. Much of the primary specification's systematic level bias was removed by a global calibration step, but the required offset differed across model and prompting configurations. Calibration improved average accuracy but did not remove all product-level error, so corrected curves still require application-specific evaluation. For this model and consumer-packaged-goods setting, the results support a limited role for digital twins as a complement to human data. The relevant question is whether bias remains stable within a defined simulation setup and decision context, rather than whether raw twin output is accurate in general.

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